Q1: 16 March (Shift 1) - Single Correct

If n is the number of irrational terms in the expansion of $\left(3^{1/4}+5^{1/8}\right)^{60}$, then $(\mathrm{n}-1)$ is divisible by:

(1) 26

(2) 30

(3) 8

(4) 7

Q2: 16 March (Shift 1) - Single Correct

Let [x] denote greatest integer less than or equal to x. If for $n \in \mathbb{N}$, $\left(1-x+x^{3}\right)^{n}=\sum_{j=0}^{3n} a_{j} x^{j}$, then $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j}+4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j}+1$ is equal to :

(1) 2

(2) 2^{n-1}

(3) 1

(4) n

Q3: 16 March (Shift 2) - Numerical

Let n be a positive integer. Let

$$\mathrm{A}=\sum_{\mathrm{k}=0}^{\mathrm{n}}(-1)^{\mathrm{k}}{ }^{\mathrm{n}} \mathrm{C}_{\mathrm{k}}\left[\left(\frac{1}{2}\right)^{\mathrm{k}}+\left(\frac{3}{4}\right)^{\mathrm{k}}+\left(\frac{7}{8}\right)^{\mathrm{k}}+\left(\frac{15}{16}\right)^{\mathrm{k}}+\left(\frac{31}{32}\right)^{\mathrm{k}}\right]$$

If 63 $\mathrm{A}=1-\frac{1}{2^{30}}$, then n is equal to ____

Q4: 17 March (Shift 1) - Single Correct

If the fourth term in the expansion of $\left(\mathrm{x}+\mathrm{x}^{\log_{2} \mathrm{x}}\right)^{7}$ is 4480, then the value of x where $x \in N$ is equal to :

(1) 2

(2) 4

(3) 3

(4) 1

Q5: 17 March (Shift 1) - Numerical

If $(2021)^{3762}$ is divided by 17, then the remainder is ____.

Q6: 17 March (Shift 2) - Single Correct

The value of $\sum_{r=0}^{6}\left({ }^{6}\mathrm{C}_{r}{ }^{-6}\mathrm{C}_{6-\mathrm{r}}\right)$ is equal to :

(1) 1124

(2) 1324

(3) 1024

(4) 924

Q7: 17 March (Shift 2) - Numerical

Let the coefficients of third, fourth and fifth terms in the expansion of $\left(x+\frac{a}{x^{2}}\right)^{n}, x \neq 0$, be in the ratio 12: 8: 3 . Then the term independent of x in the expansion, is equal to ____.

Q8: 18 March (Shift 1) - Single Correct

Let $\left(1+x+2x^{2}\right)^{20}=a_{0}+a_{1}x+a_{2}x^{2}+\ldots+a_{40}x^{40}$

then $\mathrm{a}_{1}+\mathrm{a}_{3}+\mathrm{a}_{5}+\ldots+\mathrm{a}_{37}$ is equal to

(1) $2^{20}\left(2^{20}-21\right)$

(2) $2^{19}\left(2^{20}-21\right)$

(3) $2^{19}\left(2^{20}+21\right)$

(4) $2^{20}\left(2^{20}+21\right)$

Q9: 18 March (Shift 2) - Numerical

The term independent of x in the expansion of

$\left[\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right]^{10}, x \neq 1$, is equal to ___

Q10: 18 March (Shift 2) - Numerical

Let ${}^{n}C_{r}$ denote the binomial coefficient of x^{r} in the expansion of $(1+x)^{n}$.

If $\sum_{k=0}^{10}\left(2^{2}+3k\right){}^{n}C_{k}=\alpha.3^{10}+\beta.2^{10}, \alpha, \beta \in R$,

then $\alpha+\beta$ is equal to ___

Q11: 18 March (Shift 2) - Numerical

Let P be a plane containing the line $\frac{x-1}{3}=\frac{y+6}{4}=\frac{z+5}{2}$ and parallel to the line $\frac{x-3}{4}=\frac{y-2}{-3}=\frac{z+5}{7}$. If the point $(1,-1,\alpha)$ lies on the plane P, then the value of $|5\alpha|$ is equal to ___

Answer Key

Q1 (1)	**Q2** (3)	**Q3** (6)	**Q4** (1)
Q5 (4)	**Q6** (4)	**Q7** (4)	**Q8** (2)
Q9 (210)	**Q10** (19)	**Q11** (38)	

Q1 (1)

$$\left(3^{1/4}+5^{1/8}\right)^{60}$$

$$^{60}\mathrm{C}_{\mathrm{r}}\left(3^{1/4}\right)^{60-\mathrm{r}}\cdot\left(5^{1/8}\right)^{\mathrm{r}}$$

$$^{60}\mathrm{C}_{\mathrm{r}}(3)^{\frac{60-\mathrm{r}}{4}}\cdot 5^{\frac{\mathrm{r}}{8}}$$

For rational terms.

$$\frac{r}{8}=k;0\leq r\leq 60$$
$$0\leq 8k\leq 60$$
$$0\leq k\leq\frac{60}{8}$$
$$0\leq k\leq 7.5$$

$\mathrm{k}=0,1,2,3,4,5,6,7$

$\frac{60-8\mathrm{k}}{4}$ is always divisible by 4 for all value of k Total rational terms $=8$ Total terms $=61$

irrational terms $=53$

$\mathrm{n}-1=53-1=52$

52 is divisible by 26.

Q2 (3)

$\left(1-x+x^3\right)^n=\sum_{j=0}^{3n}a_jx^j$

$\left(1-x+x^3\right)^n=a_0+a_1x+a_2x^2\ldots\ldots+a_{3n}x^{3n}$

$\sum_{j=0}^{\left[\frac{3n}{2}\right]}a_{2j}=\text{Sum of }a_0+a_2+a_4\ldots\ldots..$

$\left[\frac{3n-1}{2}\right]$

$\sum_{j=0}a_{2j}+1=\text{Sum of }a_1+a_3+a_5\ldots\ldots$

put $x=1$

$1=\mathrm{a}_0+\mathrm{a}_1+\mathrm{a}_2+\mathrm{a}_3\ldots\ldots\ldots+\mathrm{a}_{3\mathrm{n}}\quad\ldots\ldots(\mathrm{A})$

Put $x=-1$

$1=a_0-a_1+a_2-a_3\ldots\ldots\ldots+(-1)^{3n}a_{3n}\ldots\ldots$

Solving (A) and (B)

$a_0+a_2+a_4\ldots\ldots=1$

$a_1+a_3+a_5\ldots\ldots=0$

$\sum_{j=0}^{\left[\frac{3n}{2}\right]}a_{2j}+4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]}a_{2j+1}=1$

Q3 (6)

$\mathrm{A}=\sum_{k=0}^{n}{}^{n}\mathrm{C_k}\left[\left(-\frac{1}{2}\right)^2+\left(\frac{-3}{4}\right)^2+\left(\frac{-7}{8}\right)^2+\left(\frac{-15}{16}\right)^2+\left(\frac{-37}{32}\right)^2\right]$

$\mathrm{A}=\left(1-\frac{1}{2}\right)^2+\left(1-\frac{3}{4}\right)^n+\left(1-\frac{7}{8}\right)^=+\left(1-\frac{15}{16}\right)^\mathrm{n}+\left(1-\frac{31}{32}\right)^\mathrm{n}$

$\mathrm{A}=\frac{1}{2^n}+\frac{1}{4^\mathrm{n}}+\frac{1}{8^n}+\frac{1}{16^n}+\frac{1}{32^\mathrm{n}}$

$\mathrm{A}=\frac{1}{2^\mathrm{n}}\left(\frac{1-\left(\frac{1}{2^\mathrm{n}}\right)^5}{1-\frac{1}{2^\mathrm{n}}}\right)\Rightarrow\mathrm{A}=\frac{\left(1-\frac{1}{2^{5\mathrm{n}}}\right)}{(2^\mathrm{n}-1)}$

$(2^n-1)\,\mathrm{A}=1-\frac{1}{2^{5n}}$, Given $63\,\mathrm{A}=1-\frac{1}{2^{30}}$

Clearly $5\mathrm{n}=30$

$\mathrm{n}=6$

Q4 (1)

${}^7\mathrm{C_3x^4x}^{\left(3\log_2^2\right)}=4480$

$\Rightarrow\mathrm{x}^{\left(4+3\log_2^2\right)}=2^7$

$\Rightarrow\quad(4+3\mathrm{t})\mathrm{t}=7;\mathrm{t}=\log_2^\mathrm{x}$

$\Rightarrow\mathrm{t}=1,\frac{-7}{3}\Rightarrow\mathrm{x}=2$

Q5 (4)

$(2023-2)^{3762}=2023\mathrm{k_1}+2^{3762}$

$=17\mathrm{k_2}+2^{3762}(\text{ as }2023=17\times17\times9)$

$=17\mathrm{k_2}+4\times16^{940}$

$=17\mathrm{k_2}+4\times(17-1)^{940}$

$= 17k_2 + 4(17k_3 + 1)$

$= 17k + 4 \Rightarrow \text{remainder} = 4$

Q6 (4)

$$\begin{aligned}&\sum_{r=0}^{6} {}^6C_r \cdot {}^6C_{6-r}\\&= {}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \ldots\ldots + {}^6C_6 \cdot {}^6C_0\end{aligned}$$

Now,

$$\begin{aligned}&(1+x)^6(1+x)^6\\&= \left({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \ldots\ldots + {}^6C_6x^6\right)\\&\quad\left({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \ldots\ldots + {}^6C_6x^6\right)\end{aligned}$$

Comparing coefficeint of x^6 both sides

$${}^6C_0 \cdot {}^6C_6 + {}^6C_1 + {}^6C_5 + \ldots\ldots + {}^6C_6{}^{66}C_0 = {}^{12}C_6$$

$= 924$

Q7 (4)

$$\begin{aligned}T_{r+1} &= {}^nC_r(x)^{n-r}\left(\frac{a}{x^2}\right)^r\\&= {}^nC_r a^r x^{n-3r}\end{aligned}$$
$${}^nC_2a^2 : {}^nC_3a^3 : {}^nC_4a^4 = 12 : 8 : 3$$

After solving

$$n = 6, a = \frac{1}{2}$$

For term independent of 'x' $\Rightarrow$ n $=$ 3r

$$\text{r} = 2$$

$\therefore$ Coefficient is ${}^6\text{C}_2\left(\frac{1}{2}\right)^2 = \frac{15}{4}$

Nearest integer is 4

Q8 (2)

$\left(1+x+2x^2\right)^{20} = a_0 + a_1x + \ldots. + a_{40}x^{40}$ put x=1,-1

$\Rightarrow a_0 + a_1 + a_2 + \ldots. + a_{40} = 2^{20}$

$a_0 - a_1 + a_2 + \ldots. + a_{40} = 2^{20}$

$\Rightarrow a_1 + a_3 + \ldots. + a_{39} = \frac{4^{20}-2^{20}}{2}$

$\Rightarrow a_1 + a_3 + \ldots + a_{37} = 2^{39} - 2^{19} - a_{39}$

here $a_{39} = \frac{20!(2)^{19}\times 1}{19!} = 20 \times 2^{19}$

$\Rightarrow a_1 + a_3 + \ldots + a_{37} = 2^{19}\left(2^{20} - 1 - 20\right)$

$= 2^{19}\left(2^{20} - 21\right)$

Q9 (210)

$\left(\left(x^{1/3}+1\right) - \left(\frac{\sqrt{x}+1}{\sqrt{x}}\right)\right)^{10}$

$\left(x^{1/3} - x^{-1/2}\right)^{10}$

$T_{r+1} = 10C_r\left(x^{1/3}\right)^{10-r}\left(-x^{-1/2}\right)^r$

$\frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow 20 - 2r - 3r = 0$

$\Rightarrow r = 4$

$T_5 = {}^{10}C_4 = \frac{10\times 9\times 8\times 7}{4\times 3\times 2\times 1} = 210$

Q10 (19)

Instead of ${}^nC_k$ it must be ${}^{10}C_k$ i.e.

$\sum_{k=0}^{10}\left(2^2+3k\right){}^{10}C_k = \alpha.3^{10} + \beta.2^{10}$

$\mathrm{LHS}=4 \sum_{\mathrm{k}-0}^{10}{ }^{10} \mathrm{C}_{\mathrm{k}}+3 \sum_{\mathrm{k}=0}^{10} \mathrm{k} \cdot \frac{10}{\mathrm{k}} \cdot{ }^9 \mathrm{C}_{\mathrm{k}-1}$

$=4.2^{10}+3.10.2^9$

$=19.2^{10}=\alpha \cdot 3^{10}+\beta \cdot 2^{10}$

$\Rightarrow \alpha=0, \beta=19 \Rightarrow \alpha+\beta=19$

Q11 (38)

Equation of plane is $\left|\begin{array}{ccc}x-1 & y+6 & z+5 \\ 3 & 4 & 2 \\ 4 & -3 & 7\end{array}\right|=0$

, α) lies on it so Now $(1,-1,$

$\left|\begin{array}{ccc}0 & 5 & \alpha+5 \\ 3 & 4 & 2 \\ 4 & -3 & 7\end{array}\right|=0 \Rightarrow 5 \alpha+38=0 \Rightarrow 15 \alpha l=38$